

FreshMart Agentic Analytics API Documentation

AI-Powered Natural-Language Analytics Platform Orchestrated with LangGraph, Model Context Protocol (MCP), and Role-Based Access Control

API Title:	FreshMart Agentic Analytics API
API Version:	1.0.0 (OpenAPI 3.1.0)
Base URL:	<code>http://localhost:8000/api/v1</code>
Interactive Docs:	<code>http://localhost:8000/docs</code>
Specification:	<code>http://localhost:8000/openapi.json</code>
Backend Engine:	FastAPI (Python 3.12 / 3.13)
Multi-Agent Layer:	LangGraph + Ollama (Llama 3.1 & Qwen 2.5-Coder)
Database Layer:	MySQL (Analytics) & PostgreSQL (Auth/RBAC)
Tool Protocol:	Model Context Protocol (MCP)

1. Architectural Overview

FreshMart Agentic Analytics provides secure, AI-powered natural-language analytics for FreshMart enterprise business data across retail operations, customer relationship management (CRM), human resources (HRMS), inventory warehousing, purchasing, and financial performance.

Core Engineering Principles

- **Decoupled Dual-Database Architecture:** High-volume business transactional data resides in `MySQL ( company_analytics )`, strictly segregated from authentication credentials, user records, and RBAC rules stored in `PostgreSQL ( company_auth )`.
- **Controlled Database Interfacing via MCP:** AI agents never execute arbitrary SQL strings directly on connection pools. All database schema inspection and analytical queries are mediated through a read-only Model Context Protocol (MCP) tool provider with strict AST validation, 1000-row bounds, and query timeouts.
- **Backend-Authoritative RBAC:** User identity and permissions are verified by PostgreSQL JWT before natural-language parsing or query synthesis occurs. Inquiries targeting unauthorized domains are rejected with `403 Forbidden` at the API barrier.
- **LangGraph Multi-Agent Collaboration:** Specialized agent nodes collaborate sequentially to classify intent, check permissions, inspect schemas, generate read-only SQL, validate queries, plan diagnostic investigations, and synthesize grounded responses.

Interactive Documentation Links

Swagger UI: <http://localhost:8000/docs> • ReDoc: <http://localhost:8000/redoc> • OpenAPI JSON: <http://localhost:8000/openapi.json>

2. Authentication & Role-Based Access Control (RBAC)

All analytics, dashboard, administrative, and conversational endpoints enforce JWT Bearer authentication. Requests must present the HTTP header:

```
Authorization: Bearer <signed_jwt_access_token>
```

Available Test & Demo Accounts

All demo accounts are pre-seeded in the PostgreSQL authentication database with bcrypt password hashes:

Username	Password	Role	Permitted Business Domains
ceo	CeoPassword123!	CEO	Full enterprise access (Sales, CRM, Inventory, Purchases, Finance, Profit, Expenses, HR, Salaries)
sales.manager	SalesPassword123!	SALES_MANAGER	Sales orders, customer accounts, leads, product inventory catalog
finance.manager	FinancePassword123!	FINANCE_MANAGER	Financial statements, net profit margins, operating expenses, sales revenue
hr.manager	HrPassword123!	HR_MANAGER	Workforce directory, departments, employee compensation & salary records
inventory.manager	InventoryPassword123!	INVENTORY_MANAGER	Warehouse stock levels, SKU catalogs, supplier purchase orders
erp.manager	ErpPassword123!	ERP_MANAGER	Integrated purchasing, warehouse stock, sales transactions, expenses, HR
admin	AdminPassword123!	ADMIN	System administration (create users, enable/disable accounts, inspect roles)

Granular Permissions Specification (11 Total)

Permission Code	Domain	Description	Assigned Roles
VIEW_SALES	Sales	Sales orders, items, and regional revenue velocity	CEO, SALES_MGR, ERP_MGR, INV_MGR, FIN_MGR
VIEW_CRM	CRM	Customer directory, commercial leads, customer interactions	CEO, SALES_MGR
VIEW_INVENTORY	Inventory	Catalog products, stock-on-hand, safety reorder levels	CEO, SALES_MGR, ERP_MGR, INV_MGR

Permission Code	Domain	Description	Assigned Roles
VIEW_PURCHASES	Purchases	Suppliers and procurement purchase orders	CEO, ERP_MGR, INV_MGR
VIEW_FINANCE	Finance	Overall company revenue and P&L financial statements	CEO, FINANCE_MGR
VIEW_PROFIT	Finance	Net profit, operating margins, root-cause cost variance	CEO, FINANCE_MGR
VIEW_EXPENSES	Finance	Departmental expenditures and logistics costs	CEO, HR_MGR, ERP_MGR, FINANCE_MGR
VIEW_HR	HRMS	Departments, job roles, active employee headcount	CEO, HR_MGR, ERP_MGR
VIEW_EMPLOYEE_SALARY	HRMS	Confidential employee base salaries and bonus records	CEO, HR_MGR
MANAGE_USERS	Admin	Create, edit, activate, and deactivate application users	ADMIN
MANAGE_ROLES	Admin	Inspect and reassign roles and permissions	ADMIN

3. Authentication Endpoints

POST /api/v1/auth/login

Summary: Authenticate User & Generate JWT Access Token

Operation ID: loginUser

Validates user credentials against PostgreSQL bcrypt hashes. Supports both `application/json` (standard REST API) and `application/x-www-form-urlencoded` (interactive Swagger UI Authorize dialog).

Request Body (JSON):

```
{
  "username": "ceo",
  "password": "CeoPassword123!"
}
```

Response (200 OK):

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "user": {
    "user_id": 2,
    "username": "ceo",
    "email": "ceo@freshmart.local",
    "full_name": "Vikram Malhotra",
    "role": "CEO",
    "employee_id": "E006",
    "permissions": ["VIEW_HR", "VIEW_EMPLOYEE_SALARY", "VIEW_CRM", "VIEW_SALES", "VIEW_INVENTORY", "VIEW_INVENTORY"],
    "is_active": true
  }
}
```

Possible Status Codes: 200 OK, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 422 Validation Error.

GET /api/v1/auth/me

Summary: Get Current Authenticated User Profile

Operation ID: getCurrentUserProfile

Returns identity, role, and granted permissions for the authenticated bearer token holder.

Possible Status Codes: 200 OK, 401 Unauthorized, 403 Forbidden.

4. Agentic Analytics API

POST

/api/v1/analytics/query

Summary: Run Natural-Language Business Analytics Query

Operation ID: runNaturalLanguageAnalyticsQuery

Accepts a natural-language business question and executes it through the LangGraph multi-agent pipeline.

Multi-Agent Pipeline Stages:

- Request Validator:** Checks input question bounds (1–2000 characters).
- Intent Classifier:** Classifies domain (Sales, Finance, Inventory, HR, CRM), operation (AGGREGATION, COMPARISON, ROOT_CAUSE), and target tables.
- Permission Checker:** Evaluates user RBAC codes against target domains. Rejects unauthorized requests with 403 Forbidden .
- Schema Planner:** Calls MCP tool describe_table to dynamically retrieve columns and constraints.
- SQL Generator:** Qwen 2.5-Coder synthesizes precision read-only SQL.
- SQL Security Validator:** AST parser verifies query is read-only (SELECT only; blocks INSERT, UPDATE, DELETE, DROP, ALTER, TRUNCATE, OUTFILE).
- SQL Executor:** Executes validated SQL strictly through MCP with safety row limits.
- Root-Cause & Investigation Planner:** Formulates diagnostic hypotheses and computes factor variance.
- Business Analyzer & Response Generator:** Llama 3.1 synthesizes grounded executive answers with data provenance citations.

Request Schema (AnalyticsQueryRequest):

Field	Type	Required	Description
question	string	Yes	Natural-language business analytics question (1-2000 characters)
conversation_id	string	No	Optional session ID for contextual conversational continuity

Pre-configured OpenAPI Examples:

- Sales Trend:** {"question": "Show monthly sales trend for the last 12 months."}
- Inventory Alert:** {"question": "Which products are currently low in stock?"}
- Cross-Domain:** {"question": "Which products generated the highest revenue and are currently low in stock?"}
- Root-Cause Analysis:** {"question": "Why did profit decrease in August?"}
- Workforce Comp:** {"question": "Show average employee salary by department."}

Structured Response Schema (AnalyticsQueryResponse)

```

{
  "success": true,
  "question": "Why did profit decrease in August?",
  "answer": "Net profit contracted by 50.0% (-₹600,000) from ₹1,200,000 in July to ₹600,000 in August 2026",
  "data": {
    "columns": ["month_name", "total_revenue", "cogs", "operating_expenses", "net_profit", "profit_margin"],
    "rows": [
      {"month_name": "July", "total_revenue": 10500000.0, "cogs": 7100000.0, "operating_expenses": 2200000.0, "net_profit": 1200000.0, "profit_margin": 0.1143},
      {"month_name": "August", "total_revenue": 9800000.0, "cogs": 6900000.0, "operating_expenses": 2300000.0, "net_profit": 600000.0, "profit_margin": 0.0612}
    ],
    "row_count": 2,
    "truncated": false
  },
  "intent": {
    "domain": "Finance",
    "operation": "ROOT_CAUSE",
    "target_metric": "net_profit"
  },
  "sources": [
    {
      "table": "company_financials",
      "columns": ["month_name", "total_revenue", "cogs", "operating_expenses", "net_profit"],
      "purpose": "Compute monthly profit variances"
    },
    {
      "table": "expenses",
      "columns": ["expense_date", "category", "amount"],
      "purpose": "Identify emergency shipping expenditure spike"
    }
  ],
  "columns_used": ["total_revenue", "cogs", "operating_expenses", "net_profit"],
  "visualization_hint": {
    "type": "bar",
    "x_axis": "month_name",
    "y_axis": "net_profit",
    "title": "Net Profit Comparison: July vs. August 2026"
  },
  "root_cause_analysis": {
    "analysis_type": "ROOT_CAUSE",
    "summary": "Profit decreased primarily due to emergency refrigerated freight costs during monsoon surge in August 2026.",
    "period": { "current": "August 2026", "previous": "July 2026" },
    "metrics": { "profit_change_pct": -50.0, "absolute_profit_delta": -600000.0 },
    "factors": [
      {
        "factor": "Emergency Air Freight Logistics Surge",
        "previous_value": 89326.0,
        "current_value": 430000.0,
        "change": 340674.0,
        "change_pct": 381.4,
        "impact": "negative",
        "confidence": "HIGH",
        "source": ["expenses"]
      }
    ],
    "confidence": "HIGH"
  },
  "recommendations": [

```

```
{
  "title": "Institute Emergency Logistics Surcharge Buffers",
  "reason": "Uncontracted spot-market freight costs eroded 50% of monthly operating margin.",
  "priority": "HIGH",
  "related_domain": "Finance",
  "suggested_action": "Secure secondary contracted regional logistics carriers with fixed surge rate"
}
],
"confidence": "HIGH",
"user_role": "CEO",
"user_name": "Vikram Malhotra",
"timestamp": "2026-09-25T15:20:00Z"
}
```

5. Executive Dashboard & Telemetry APIs

GET /api/v1/analytics/dashboard

Summary: Get Executive Dashboard KPIs & Sales Velocity Trends

Operation ID: getExecutiveDashboardMetrics

Returns deterministic, real-time executive dashboard metrics queried directly from FreshMart MySQL.

Query Parameters: range (string, enum: today , 7d , 30d , 90d , 12m , default: 12m)

Response Structure (DashboardResponse):

- kpis : Top-line cards (Total Revenue: ₹14.82Cr, Customers: 1,250, Orders: 14,820, Active Employees: 500).
- sales_trend : 12-month historical monthly revenue and order volumes.
- category_distribution : Sales distribution by category (Fresh Produce, Dairy & Eggs, Snacks, etc.).
- top_products : High-velocity SKUs with active warehouse balances.
- insights : Live telemetry alerts (inventory reorder warnings, August profit anomaly).

GET /api/v1/analytics/insights

Summary: Get Enterprise Insights, Anomalies & Strategic Recommendations

Operation ID: getEnterpriseInsightsAndAnomalies

Returns structured, grounded insights and detected statistical anomalies calculated from ledger telemetry.

Response Structure (EnterpriseInsightsResponse):

- insights : Enterprise cards with headline metric, percentage variance, evidence table, and recommendation.
- anomalies : Statistical outlier records (e.g. August 2026 Emergency Shipping Expense departing +381.4% from baseline).

6. Administration APIs (User & Role Management)

Administrative endpoints require `MANAGE_USERS` or `MANAGE_ROLES` permission, granted exclusively to the `ADMIN` role.

GET `/api/v1/admin/users`

Operation ID: `listAllApplicationUsers` | **Permission:** `MANAGE_USERS`

Lists all application user accounts, assigned roles, permissions, employee linkages, and active status.

POST `/api/v1/admin/users`

Operation ID: `createApplicationUser` | **Permission:** `MANAGE_USERS`

Provisions a new user account with bcrypt password hashing. Returns `201 Created`.

```
{
  "username": "anita.finance",
  "email": "anita.f@freshmart.local",
  "password": "SecurePass2026!",
  "full_name": "Anita Sharma",
  "role_id": 7,
  "employee_id": "E010"
}
```

PATCH `/api/v1/admin/users/{user_id}/status`

Operation ID: `updateUserAccountStatus` | **Permission:** `MANAGE_USERS`

Enables or disables (deactivates) a user account. Deactivated accounts are immediately blocked.

```
{ "is_active": false }
```

PATCH `/api/v1/admin/users/{user_id}/role`

Operation ID: `updateUserApplicationRole` | **Permission:** `MANAGE_ROLES`

Reassigns a user's application role, updating their granted permissions dynamically.

```
{ "role_id": 5 }
```

GET /api/v1/admin/roles

Operation ID: listApplicationRolesAndPermissions | **Permission:** MANAGE_ROLES

Returns all 7 application roles and their complete assigned permission codes.

7. AI Chatbot & Health Diagnostic APIs

POST /api/v1/chat/message

Summary: Send Conversational Business Inquiry

Operation ID: sendChatMessage

Role-scoped conversational AI chat endpoint powered by Ollama Llama 3.1 8B. Injects authorized business domains into the system prompt.

```
{
  "message": "What business domains am I authorized to query?",
  "conversation_id": "conv-001"
}
```

GET /health

Summary: API Health Check (Liveness Probe)

Operation ID: getLivenessHealthCheck

```
{ "status": "ok" }
```

GET /api/v1/health

Summary: Detailed System & Services Diagnostic Check

Operation ID: getDetailedSystemHealthCheck

Verifies operational status of FastAPI, Ollama LLM models, MySQL, and PostgreSQL.

```
{
  "status": "healthy",
  "python_version": "3.13.15",
  "services": {
    "fastapi": { "status": "online" },
    "ollama": {
      "status": "online",
      "available_models": ["llama3.1:8b", "qwen2.5-coder:7b"],
      "required_sql_model": "qwen2.5-coder:7b",
      "required_agent_model": "llama3.1:8b",
      "sql_model_ready": true,
      "agent_model_ready": true
    },
    "mysql": { "status": "online", "host": "localhost:3306", "version": "8.0.36" },
    "postgresql": { "status": "online", "host": "localhost:5432", "version": "PostgreSQL 16.1" }
  }
}
```

8. Error Handling & Integration Code Samples

HTTP Status Code Reference

Status Code	Reason	Description
200 OK	Success	Request completed and response payload returned.
201 Created	Resource Created	User account successfully provisioned.
400 Bad Request	Client Error	Missing parameters, empty question, or duplicate username/email.
401 Unauthorized	Authentication Failure	Missing, invalid, or expired JWT bearer token.
403 Forbidden	Authorization Failure	User lacks the required RBAC domain permission.
404 Not Found	Resource Not Found	Target user or role ID does not exist in the database.
422 Unprocessable	Validation Error	Request payload failed Pydantic schema validation.
500 Internal Error	Server Error	Multi-agent execution failure or database error.
503 Unavailable	Service Offline	Ollama LLM or backend service unreachable.

Python Integration Example

```
import requests

BASE_URL = "http://localhost:8000/api/v1"

# 1. Authenticate
login_resp = requests.post(f"{BASE_URL}/auth/login", json={
    "username": "ceo",
    "password": "CeoPassword123!"
})
token = login_resp.json()["access_token"]
headers = {"Authorization": f"Bearer {token}"}

# 2. Run Natural-Language Analytics Query
query_resp = requests.post(
    f"{BASE_URL}/analytics/query",
    headers=headers,
    json={"question": "Why did profit decrease in August?"}
)
result = query_resp.json()

print(f"Question: {result['question']}")
print(f"Answer: {result['answer']}")
print(f"Factors: {len(result.get('root_cause_analysis', {}).get('factors', []))} causal factors identified")
```

cURL Command Example

```
curl -X POST "http://localhost:8000/api/v1/analytics/query" -H "Authorization: Bearer <YOUR_JWT_TOKEN>"
```

FreshMart Agentic Analytics Platform • Confidential & Enterprise Proprietary • Powered by FastAPI, LangGraph, Ollama, & MCP.